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ALICE L WALTON SCHOOL OF MEDICINE

Pericardial Effusion and Cardiac Tamponade

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Adapted from: Pericardial Effusion and Cardiac Tamponade

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Learning Objectives (4) Versions

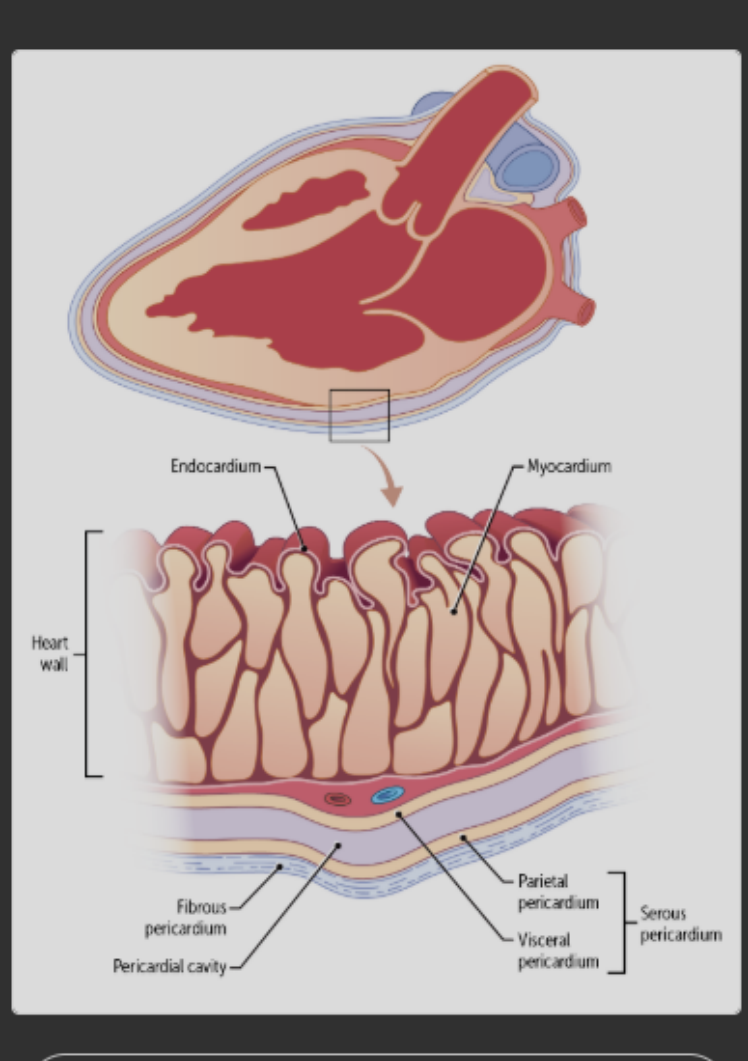
After completing this brick, you will be able to:

1. Describe the clinical presentations of pericardial effusion and cardiac tamponade
2. Explain the pathophysiology of pericardial effusion and cardiac tamponade.
3. Interpret the … al effusion and

may contain more than 2000 mL of fluid! When a pericardial effusion gets so large that it starts compressing the heart and interfering with its ability to pump blood, we refer to it as **cardiac tamponade**. Cardiac tamponade is a medical emergency, and, if left untreated, can progress to obstructive shock and death.

Pericardial Anatomy

Before delving into more detail, let's briefly review some anatomy so that we can better visualize these disorders. Take a look at Figure 1.



may instead relate to the underlying cause of the pericardial effusion, such as fever or pleuritic chest pain in acute pericarditis or weight loss in cancer.

However, patients presenting with cardiac tamponade will appear ill and have signs and symptoms of **hemodynamic compromise**. Their blood pressure might be low and their jugular veins dilated, and they may also complain of dizziness and fatigue.

What Is the Pathophysiology of Pericardial Effusion and Cardiac Tamponade? (LO#2)

Pericardial effusions and cardiac tamponade are due to any process that leads to fluid from the bloodstream entering the pericardial space.

Causes of Pericardial Effusion

Pericardial effusions can be caused by almost any disease process that affects the pericardium:

- **Acute pericarditis** [eg, from viral infection, autoimmune disease, myocardial infarction, blunt/sharp trauma, uremia (symptomatic renal failure), ionizing radiation]—pericardial inflammation causes increased capillary permeability, resulting in effusion.
- **Nephrotic syndrome or liver failure**—decreased plasma oncotic pressure results in pericardial effusion as fluid moves from the blood into the interstitial space.
- **Malignancy**—usually caused by metastatic spread to the pericardium from other sites (malignant effusion).
- **Cardiac rupture** … trating trauma) or

physical exam with the results of electrocardiography (ECG), radiologic studies, and lab tests.

History

Small, chronic pericardial effusions are often discovered incidentally. In contrast, **rapidly developing pericardial effusions are high risk for tamponade** and hemodynamic compromise (eg, dizziness, dyspnea, or syncope due to low cardiac output). Other patient-history findings may help us identify the specific cause that led to the pericardial effusion (eg, malignancy, myocardial infarction, uremia). Chronic effusions can result in tamponade if they reach the limit of pericardial stretching.

What characteristic of a pericardial effusion is most important regarding the likelihood of developing symptoms?

Physical Examinati…

Patients with cardiac tamponade may also develop **pulsus paradoxus**, which is a decrease in systolic blood pressure by >10 mm Hg during inspiration. Deep inspiration decreases intrathoracic pressure, which increases venous return. In tamponade, the low pressure right side of the heart is unable to expand, forcing the interventricular septum to the left. Left ventricular size decreases, decreasing stroke volume, cardiac output, and blood pressure.

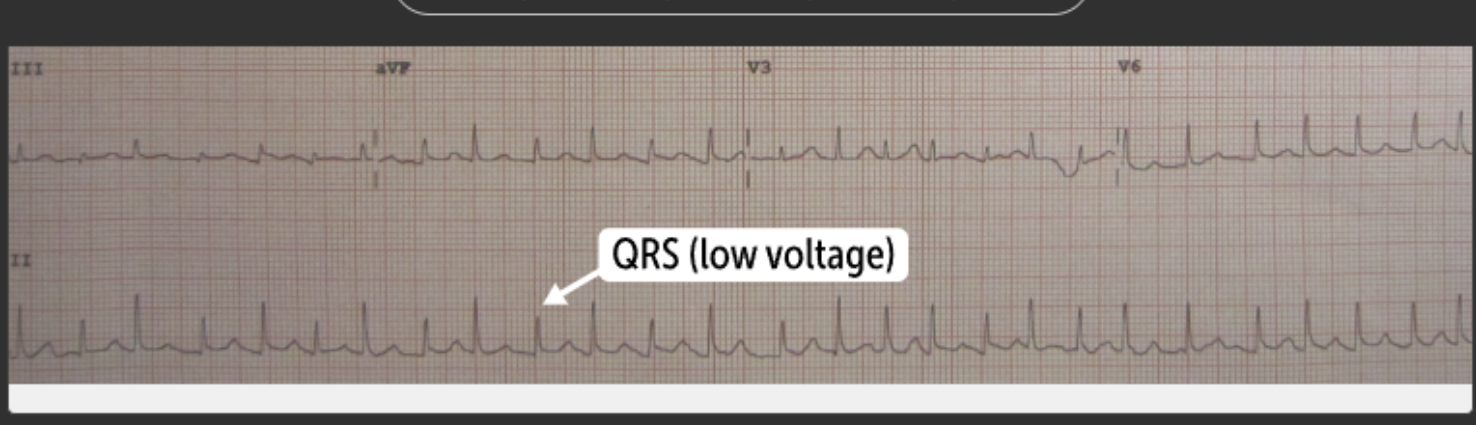
CLINICAL CORRELATION

Measuring pulsus paradoxus involves the following steps:

1. Inflate the blood pressure (BP) cuff until Korotkoff sounds are no longer heard.
2. Slowly deflate the BP cuff until sounds are heard intermittently (rather than with every heartbeat), and note the BP at this point. This corresponds to the higher systemic BP that occurs during expiration.
3. Continue to slowly deflate the cuff until sounds are heard with every beat, during inspiration and expiration, and note the BP at this point.
4. If the difference between the pressures noted during step 2 and step 3 is >10 mm Hg, this is considered pulsus paradoxus.

The original "paradox" meant that the radial pulse disappears during inspiration even though regular heart sounds are still auscultated with a stethoscope.

Notably, the Kussmaul sign (an increase in jugular venous pressure during inspiration) is usually not seen in cardiac tamponade and is instead a finding typical of constrictive …



QUIZ

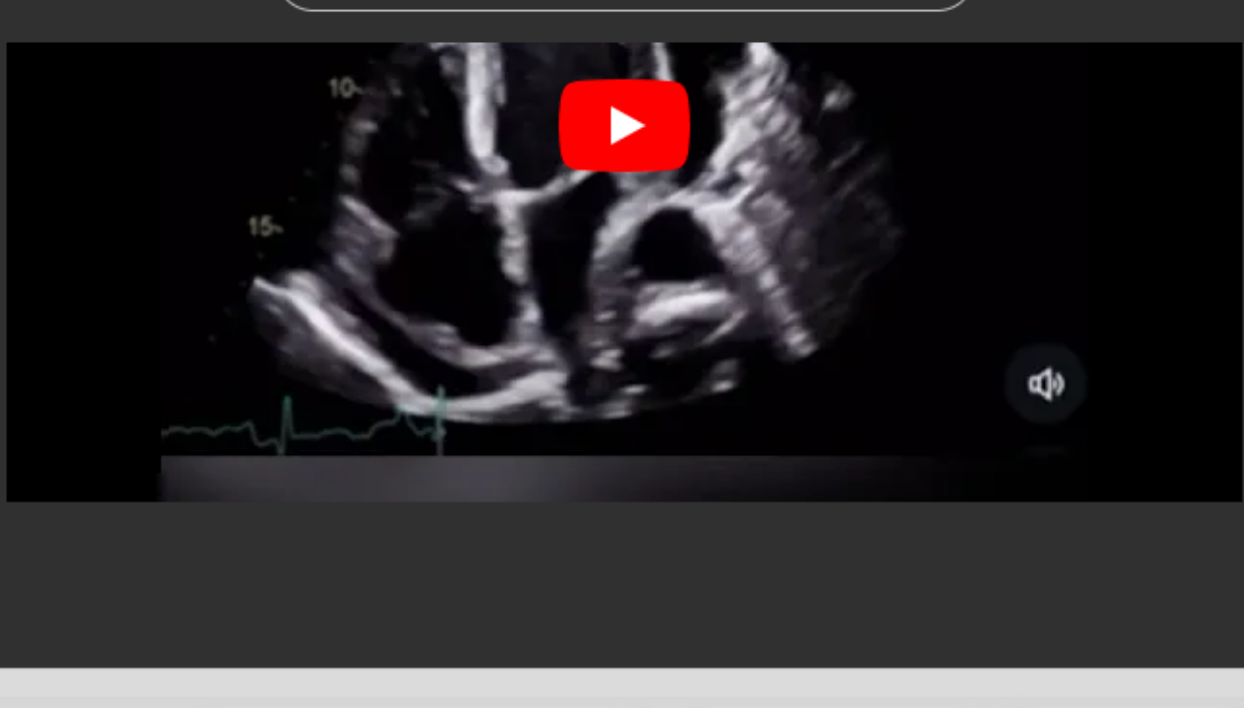
Figure 4. Electrical alternans

This sign is produced by the "swinging" movement of the heart in the fluid-filled pericardial sac. However, it's only seen in a minority of patients with tamponade (20%–25%).

ECG findings more commonly include sinus tachycardia and low-voltage QRS complexes in all leads. The latter occurs because the fluid around the heart limits electric conduction to the ECG leads. However, these findings are less specific.

Radiology Studies

A chest x-ray of a patient with pericardial effusion may reveal an enlarged heart silhouette with a "water bottle", rounded shape (Figure 5). Compare this to cardiomegaly, where the typical shape of the cardiac shadow is maintained. Note that to be detected by chest x-ray, at least 250 mL of fluid must be present in the pericardial cav… noticed.



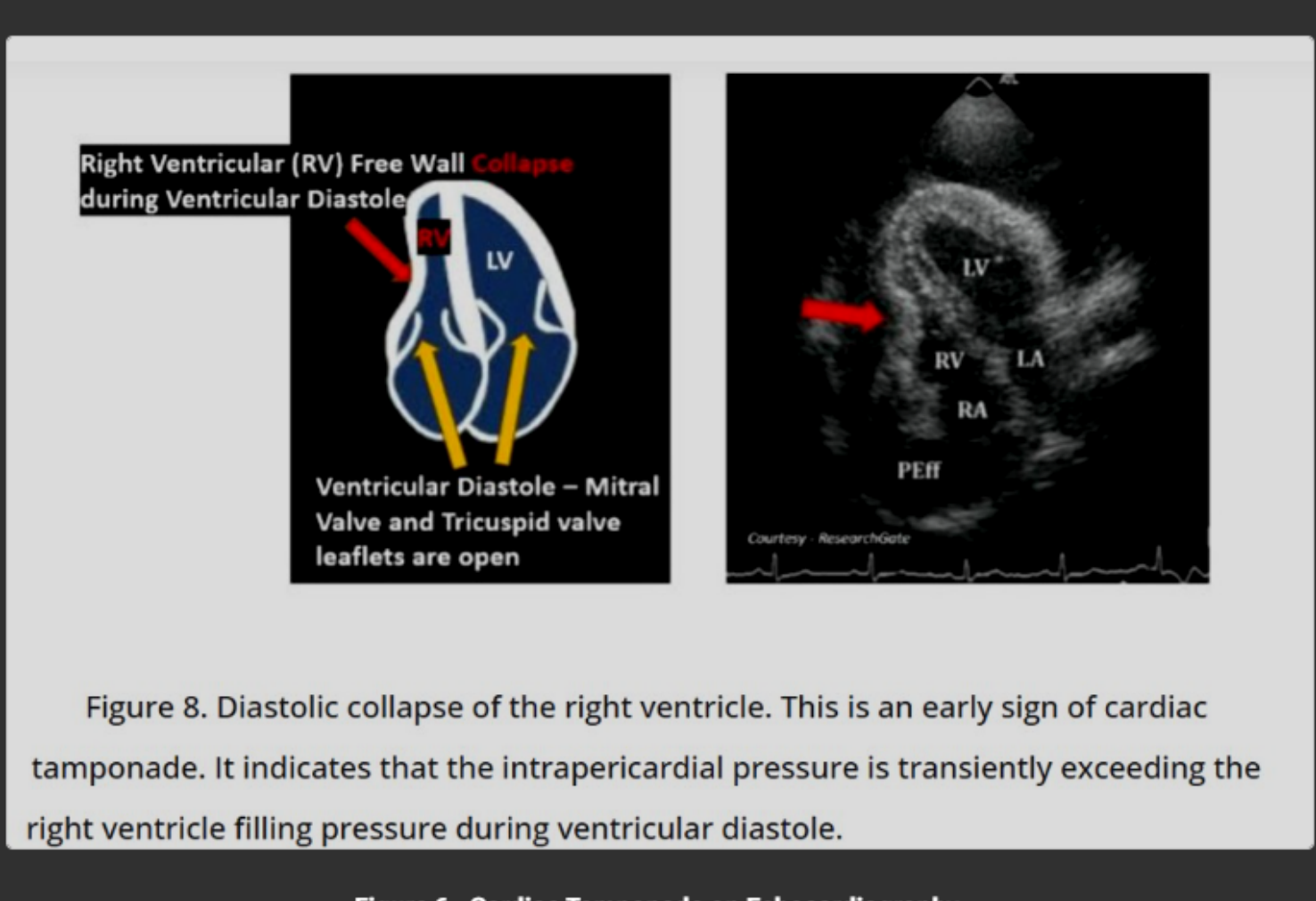
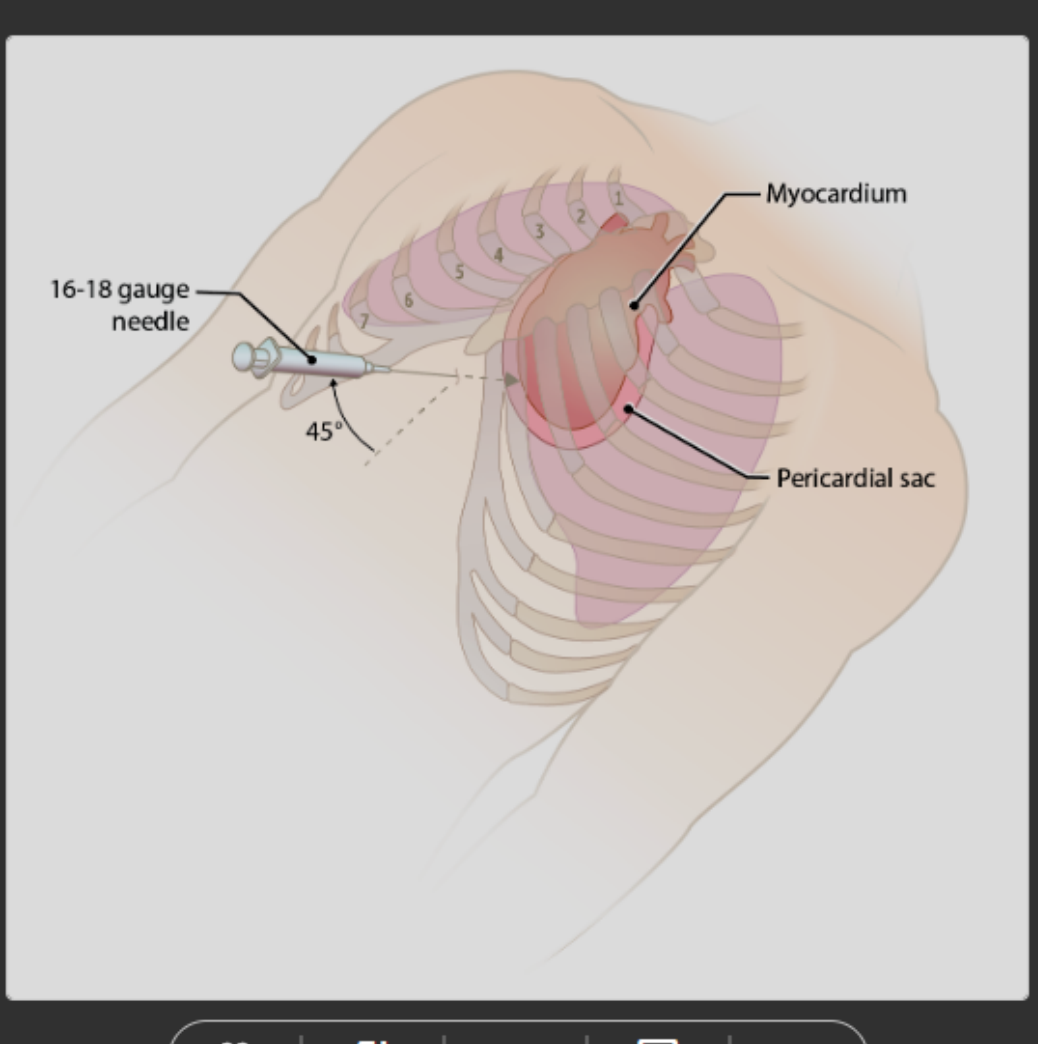

Right Ventricular (RV) Free Wall Collapse during Ventricular Diastole
RV
LV
Ventricular Diastole – Mitral Valve and Tricuspid valve leaflets are open
LV
RV
LA
RA
PEff
Courtesy · ResearchGate

Figure 8. Diastolic collapse of the right ventricle. This is an early sign of cardiac tamponade. It indicates that the intrapericardial pressure is transiently exceeding the right ventricle filling pressure during ventricular diastole.

Figure 6. Cardiac Tamponade on Echocardiography.

If a patient with pericardial effusion is hemodynamically stable with no evidence of cardiac tamponade, it is not necessary to drain the effusion. Treatment of the underlying condition may lead to clinical improvement or resolution without the need for drainage.

But in patients with cardiac tamponade and hemodynamic compromise, emergent drainage of the pericardial effusion is required. This is best achieved through **pericardiocentesis**, which involves the removal of fluid from the pericardial space via percutaneous (ie, through the skin) needle aspiration (Figure 7).



tamponade is suspected. A bedside echocardiogram is ordered stat and confirms the diagnosis. MR undergoes emergency surgery to drain the pericardial fluid and repair her other injuries.

Summary

- Pericardial effusion is an accumulation of fluid in the pericardial space, which can be asymptomatic if small but may lead to cardiac tamponade if large.
- Cardiac tamponade occurs when pericardial effusion impairs right ventricular filling, leading to hemodynamic compromise and requiring urgent intervention.
- With tamponade, increased RV volume during inspiration can push the interventricular septum towards the left ventricle, further decreasing cardiac output resulting in pulsus paradoxus.
- Causes of pericardial effusion include acute pericarditis, nephrotic syndrome, liver failure, malignancy, cardiac rupture, aortic dissection, and hypothyroidism.
- Beck triad (jugular venous distention, muffled heart sounds, and hypotension) is indicative of cardiac tamponade, along with pulsus paradoxus and electrical alternans on ECG.
- Echocardiography is the preferred imaging modality for diagnosing pericardial effusion and cardiac tamponade.
- Pericardiocentesis is the treatment of choice for cardiac tamponade, and a pericardial window may be used for continuous drainage in recurrent cases.

2. Which of the following is the most important factor in distinguishing cardiac tamponade from a nontamponade pericardial effusion?

A. ○ Flattened neck veins
B. ○ Displaced point of maximal impulse
C. ● Hypotension
D. ○ Hypertension
E. ○ Increased white blood cell count

Hide Explanation

The correct answer is hypotension, a sign of hemodynamic compromise characteristic of tamponade (C). As fluid continues to accumulate within the pericardial cavity, pressure in the pericardial space increases and compromises the filling of the right side of the heart. This reduces preload and left ventricle cardiac output. Cardiac tamponade can present with the Beck triad of distended (not flattened) neck veins (A), hypotension (not hypertension) (D), and muffled heart sounds. Neither a displaced point of maximal impulse (B) nor an increased white blood cell count are necessary for a diagnosis of cardiac tamponade (E).

3. A patient has established cardiac tamponade; what is the best next step in their management…

B. ○ Myocardial infarction
C. ○ Hypothyroidism
D. ● Osteoporosis
E. ○ Uremia

Hide Explanation

The correct answer is osteoporosis (D). Cardiothoracic surgery (A), myocardial infarction, (B), hypothyroidism (C), and uremia (E) have all been associated with increased risk for pericardial effusion.

How would you rate this Brick?

References

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length alternans. *Zhonghua Minguo Xin Zang Xue Hui Za Zhi.* 2013;29(2):192-195.
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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Describe the clinical presentations of pericardial effusion and cardiac tamponade~~
02 ☑ ~~Explain the pathophysiology of pericardial effusion and cardiac tamponade.~~
03 ☑ ~~Interpret the clinical findings to diagnose pericardial effusion and cardiac tamponade~~
04 ☑ ~~Outline the treatment plan for pericardial effusion and cardiac tamponade~~

Objective Confidence: 4/4